RTL Vibecoding: Spec format and LLM RTL: question and success criteria

Project 11 • Luke Summers • November 12, 2026

Does the form of a hardware spec (prose, table, timing diagram, or all three) change how often an LLM writes correct RTL for the same design?

WHY IT MATTERS

Small FPGA blocks in embedded sensor pipelines (SPI receiver, FIFO, encoder) are specified in datasheets as tables and waveforms, but LLMs are usually prompted in prose.

Constraint: functional and cycle-accurate timing correctness.

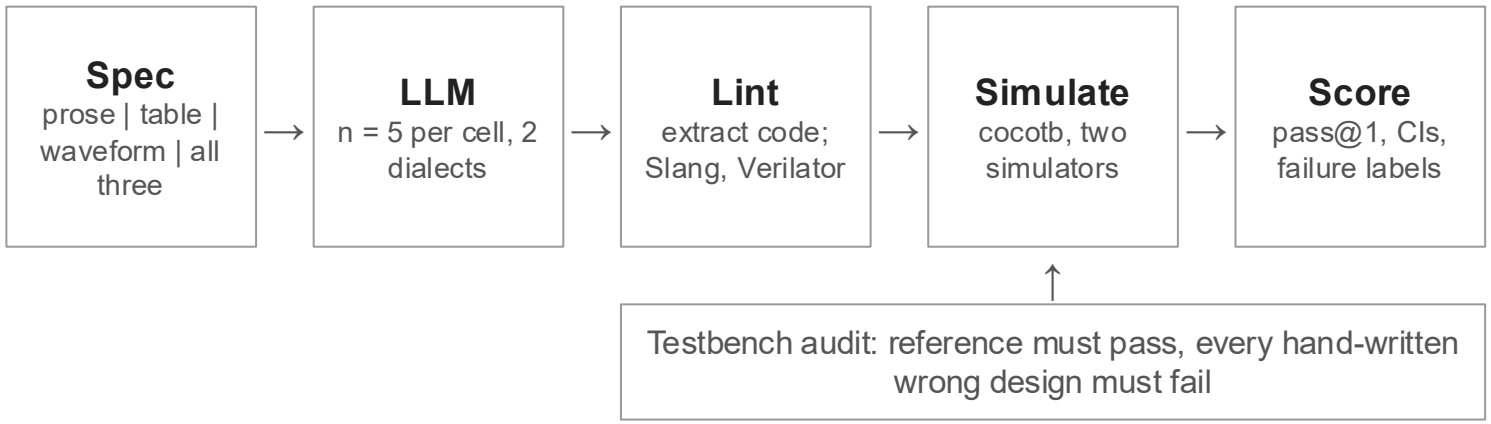
SUCCESS CRITERIA

Baseline: prose spec of the same design, graded by the same testbench.

Metric: pass@1 in RTL simulation per format, with 95% confidence intervals.

Success: a format difference whose paired CI excludes 0, or a null with CI half-width ≤ 0.10 on problems that are not at ceiling.

Approach and current implementation



TEST SETTING
MacBook; Icarus 13, Verilator 5.054, cocotb 2.1
3 problems × 4 formats × 2 dialects; Claude Haiku, Sonnet, Opus 5.5
RTL simulation only: no hardware, no live sensing

TEAM CONTRIBUTION
Built: specs, references, testbenches, 17 mutants, pipeline scripts
Reused: simulators, cocotb, Slang, pass@k estimator

TESTED: pipeline on 270 real generations (3 problems, 3 models)
PLANNED: 9 more problems (NVIDIA CVDP, RTLLM, own), a weaker model

Key choice: every format of a problem is graded by one testbench, and that testbench must first prove it catches plausible wrong designs.

Preliminary evidence and what it means

passes / generations	prose	table	waveform	combined
Claude Haiku 5.5	30/30	**29/30**	15/15	15/15
Claude Sonnet 5.5	30/30	30/30	15/15	15/15
Claude Opus 5.5	30/30	30/30	15/15	15/15

pass@1 counts in RTL simulation: 3 problems (SPI receiver, priority encoder, FIFO), 5 samples per cell; prose and table also in Verilog-2001.

Testbench audit: 12/12 reference runs pass, 17/17 wrong designs fail, on both simulators.

OBSERVATION
269 of 270 generations pass. The one failure: Haiku, SPI receiver, table spec (bit 7 lost).
14 SystemVerilog designs pass on Verilator but do not compile on Icarus.

INTERPRETATION
The pipeline and testbenches work on real model output. It shows nothing about format: these problems are at ceiling.

LIMITATION
3 easy problems, one model family. Open: will harder problems leave headroom?

Source: runs/pilot (270 generations, Nov 10, 2026), checkpoint2/evidence_log.txt, github.com/lukesummers-20/202a-proj. Measured in simulation; temperature at API default.

Evaluation plan and next steps

EVALUATION
Baseline: prose spec. Alternatives: table, waveform, combined.
Metric: pass@1 on 12 audited problems (3 own, 5 NVIDIA CVDP, 3 RTLLM, 1 new), 10 samples per cell, 3 Claude models + 1 weaker model.
Ablation: Verilog-2001 vs SystemVerilog on prose and table.

MAIN RISK
Harder problems are still at ceiling. Fallback: answer the question on weaker models and report the ceiling for frontier models as a result.

NEXT STEPS
Port 3 external problems, run difficulty screen — Luke — Nov 17
Reach 12 audited problems — Luke — Nov 24
Full runs and failure labels — Luke — Dec 1

SCOPE
Minimum: 12 audited problems × 4 formats × 3 models, pass@1 table with CIs, and a labelled failure taxonomy.

FEEDBACK REQUESTED: If frontier models stay at ceiling, is a format effect shown only on a weaker model an acceptable answer, or should I make the specs harder instead?